1-Day Technical Challenge: MERN Task Manager

Role: MERN Stack Developer
Time Limit: 1 Day

1. Objective
Build a simple **Task Management App**.
Admin: Can create users and assign tasks to them.
Developer: Can see their assigned tasks and mark them as "Completed".

2. Tech Stack
Frontend: React.js, Tailwind CSS
Backend: Node.js, Express.js
Database: MongoDB, Mongoose
Validation: Joi,Zod

Phase 1: The Backend (API)
Keep it simple. Focus on getting the data to flow.

A. Database Models (Schema)
You only need two schemas:

1. **User:** `name`, `email`, `password`, `role` ('admin' or 'dev').
2. **Task:** `title`, `description`, `dueDate`, `status` ('pending', 'completed'), `assignedTo` (User ID).

B. Required API Endpoints
Do not worry about versioning (`/v1`) for now.

1. Auth

`POST /register`: Create a new user.
`POST /login`: Authenticate and return a token/JWT.

2. Users
`GET /users`: **(Admin Only)** Returns a list of all users.
Why: The Admin needs this list to populate a dropdown menu when assigning a task to a specific developer.

3. Tasks

`POST /tasks`: **(Admin Only)** Create a task and assign it to a user ID.

`GET /tasks`: Get tasks based on role.

If Admin: Returns **all** tasks.

If Dev: Returns **only tasks assigned to me**.

`PUT /tasks/:id`: **(Dev Only)** Update status to 'Completed'.

`DELETE /tasks/:id`: **(Admin Only)** Delete a task.

Why: Allows the Admin to remove tasks created by mistake or cancelled work.

Phase 2: The Frontend (React)

Don't spend too much time on design. Use Tailwind to make it clean, but functionality comes first.

Required Pages

1. Login / Register

Simple form to log in.

Save the JWT token in `localStorage` to manage sessions.

2. Dashboard (The Main View)

If logged in as Admin:

Create Task: A form with a dropdown to select a Developer (populated by `GET /users`).

All Tasks: Show a list of all tasks.

Delete: A button to delete a task.

If logged in as Developer:

My Tasks: Show a list of tasks assigned only to the logged-in user.

Complete: A button on each task to mark it as "Completed".

3. Submission Instructions

Priorities:

1. **Functionality:** Can I log in? Can I create a task? Does the Developer see *only* their own tasks?
2. **Code Cleanliness:** Is the code easy to read? (Use meaningful variable names).

Deliverables:

A GitHub link with the code.

A short `README.md` telling us how to start the app (e.g., `npm run dev`).

Best of Luck!